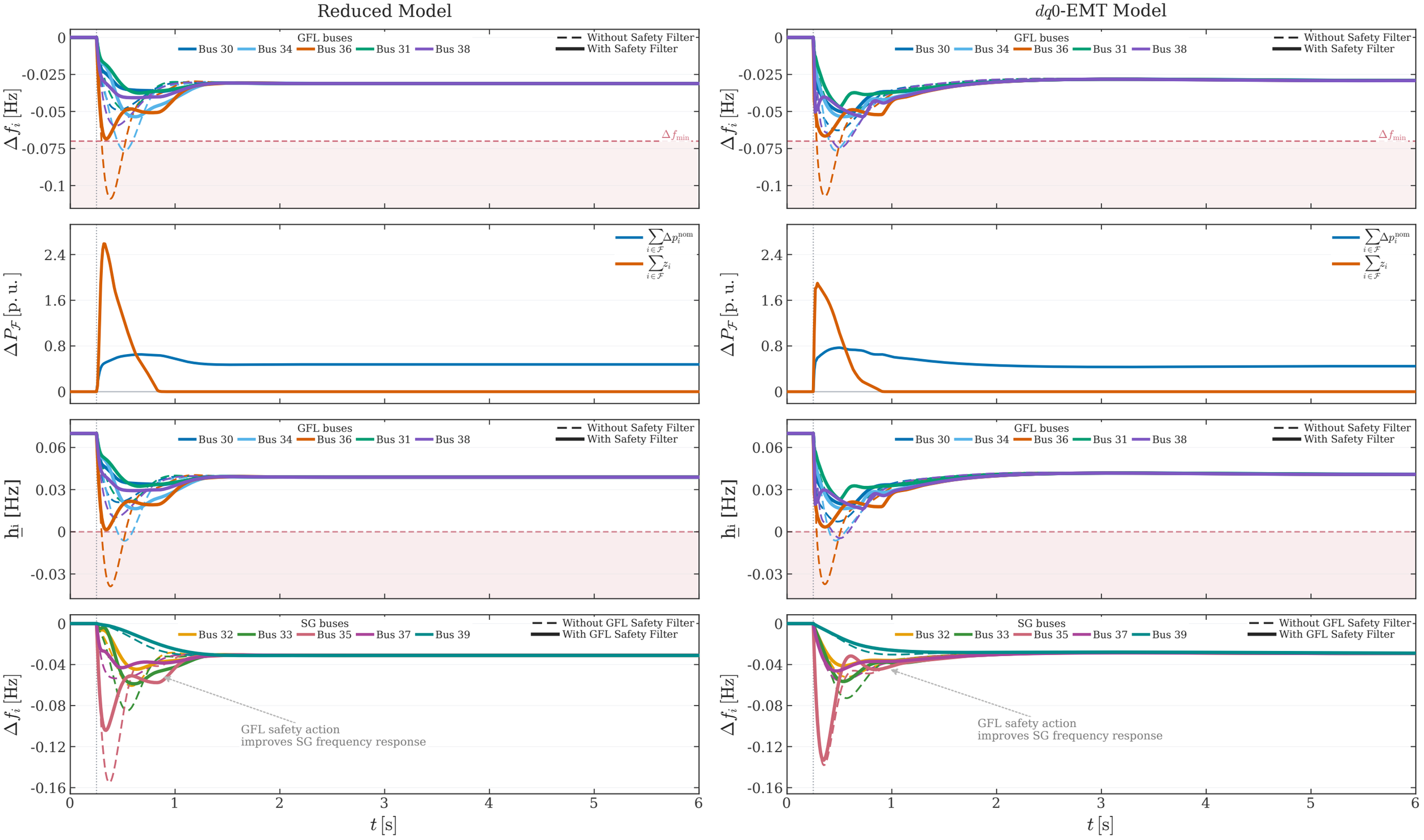

Reduced Model
dq0-EMT Model
GFL buses
Bus 30
Bus 34
Bus 36
Bus 31
Bus 38
Without Safety Filter
With Safety Filter
Δf_i [Hz]
$\Delta f_{\min}$
$\sum_{i\in\mathcal{F}} \Delta p_i^{\text{nom}}$
$\sum_{i\in\mathcal{F}} z_i$
$\Delta P_{\mathcal{F}}$ [p. u.]
$\underline{h}_i$ [Hz]
SG buses
Bus 32
Bus 33
Bus 35
Bus 37
Bus 39
Without GFL Safety Filter
With GFL Safety Filter
GFL safety action
improves SG frequency response
t [s]